Physical & Chemical Changes

Exam code: 0620 & 0971

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Physical & chemical changes

Physical changes

- Physical changes do not produce any new chemical substances
- These changes are often:
  - **Easy to reverse**
  - Relatively **easy to separate**
- Examples of physical changes include:
  - Changing state, e.g. melting / solid → liquid
  - Making a mixture from 2 or more substances
  - Dissolving a solute in a solvent

Chemical changes

- During chemical changes / reactions, **new chemical** substances are formed that have very different **properties** to the reactants
- Most chemical changes are **difficult to reverse**
- There may be signs that a new substance has formed, such as:
  - Colour changes
  - Temperature changes
  - Effervescence (fizzing)

Colour change

- One example of a reaction that shows a colour change is the metal displacement reaction of silver and copper
  - Orange-brown copper metal is added to a colourless solution of silver nitrate
  - As the reaction proceeds the copper displaces the silver from the solution
  - This causes two colour changes:
    1. The solid inside the beaker changes from orange-brown to silver
    2. The solution changes from colourless to blue

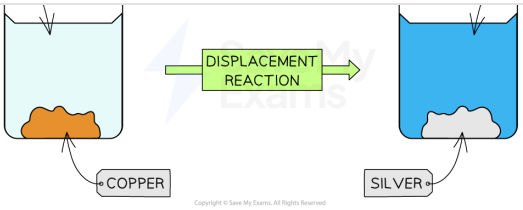
The metal displacement reaction of silver and copper

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



The reaction shows two colour changes; one in the colour of the solution and one in the colour of the solid

- Another reaction, from the syllabus, that shows a clear colour change is the halide ion test
  - From colourless to cream / white / yellow depending on the halide ion
  - This also includes a change of state as a solid / precipitate is formed

Temperature change

- Chemical reactions that give a **temperature change** can be grouped into two categories:
  - Those that give out heat / thermal energy - **exothermic**
  - Those that take in heat / thermal energy - **endothermic**

Exothermic reactions

- Some examples of reactions that cause an **increase in temperature** are:
  - The reaction of calcium oxide with water to form calcium hydroxide is highly exothermic
  - The reaction of sodium with water is exothermic and effervesces

Endothermic reactions

- Reactions that cause a **decrease in temperature** are less commonly talked about than exothermic reactions
- Examples include:
  - Photosynthesis
    - Light energy is absorbed during the process of converting carbon dioxide and water into glucose and oxygen
  - When solid ammonium chloride is dissolved in water
    - Heat / thermal energy is absorbed from the surroundings, causing the temperature to decrease
    - This reaction is commonly used in cold packs

Effervescence

- Effervescence, or fizzing, is another sign of a chemical reaction
- Chemical reactions that cause effervescence often involve acids:
  - The reaction of alkali metals with water
    - The reaction of the alkali metals, such as sodium, with water releases hydrogen gas
    - Metal + water → metal hydroxide + hydrogen

- A change in pH

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